

Multiple Choice (10 marks)

1. Elements with partially filled 4 f or 5 f orbitals include all of the following EXCEPT
 - (a) Eu
 - (b) Gd
 - (c) Am
 - (d) Cu
2. Assuming complete neutralization, calculate the number of milliliters of 0.025 M H_3PO_4 required to neutralize 25 ml of 0.030 M $\text{Ca}(\text{OH})_2$.
 - (a) 10 ml
 - (b) 30 ml
 - (c) 55 ml
 - (d) 40 ml
 - (e) 25 ml
3. A wine has an acetic acid (CH_3COOH , 60 g/formula weight) content of 0.66% by weight. If the density of the wine is 1.11 g/ml, what is the formality of the acid?
 - (a) $1.6 \times 10^{-3}F$
 - (a) $1.5 \times 10^{-3}F$
 - (a) $1.4 \times 10^{-3}F$
 - (a) $0.8 \times 10^{-3}F$
 - (a) $0.9 \times 10^{-3}F$
4. Under standard temperature and pressure conditions, compare the relative rates at which inert gases, Ar, He, and Kr diffuse through a common orifice.
 - (a) .1002 : .3002 : .4002
 - (b) .3582 : .4582 : .0582
 - (c) .2582 : .4998 : .3092
 - (d) .1582 : .6008 : .2092
 - (e) .1582 : .4998 : .1092
5. Would you expect He_2^+ to be more stable than He_2 ? Than He ?

6. He_2^+ is the most stable among He , He_2 , and He_2^+
6. He_2^+ is more stable than He_2 but less stable than He
6. He , He_2 , and He_2^+ have the same stability
6. He_2^+ is the least stable among He , He_2 , and He_2^+
6. He and He_2 have the same stability, both more stable than He_2^+

True / False (10 marks)

Answer True or False

6. True or False: The ionization energy of singly ionized helium calculated using the Bohr theory is 52.307 eV.
7. True or False: At 12.19 K, a pure sample of an ideal gas can exhibit a pressure of 1 atm and a concentration of 1 mole liter⁻¹.
7. True or False: NiSO_3 is expected to have low solubility in water due to its ionic nature and the limited solubility of sulfites in aqueous solutions.
8. True or False: The missing mass in ${}^{19}_9\text{F}$ is 0.00549 amu, which suggests a significant mass loss due to
8. True or False: The molality of a solution containing 49 g of H_2SO_4 dissolved in 250 g of water is 2.0 moles/kg.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of diluting a 90% ammonia solution to achieve a 60% concentration. How much water is required, and what calculations support your answer?
12. Discuss the thermodynamic principles governing the expansion of one cubic foot of air when heated from 0 to 546,000 and calculate the resulting volume.

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